

ARIM — Autonomous Reliability Integrity Module

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OriginID: OOF-OID-AI-ARIM-2026-06-03-0001

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: Cognitive Governance Intelligence Architecture (CLIA®)

Operational Layer: Agent Cognition Governance Layer

Governed Space: Autonomous Reliability Integrity

Category: AI & Interpretation

Subcategory: Autonomous Cognition Reliability Governance

Type: Agent Cognition Integrity Module

Parent Standard: Agent Cognition Integrity Standard (ACIS)

Version: 1.0

Status: Canonical · Open Module

Origin Date: 3 June 2026

Compatibility: OOF Methodology OS ·

Agent Cognition Integrity Standard (ACIS) ·

Cognitive Integrity Standard (CIS) ·

Runtime Integrity Standard (RIS) ·

Operational Decision Integrity Standard (ODIS) ·

Operational Constraint Integrity Standard (OCNS) ·

INTEGROS® — Integrity Standard ·

Ethical Virtual Integrity Protocol (EVIP)

AI-Readable: Yes

Authority: OOF

Protection: MIP — Methodological Intellectual Property

Canonical Language: English (UCL)

Canonical Definition

Autonomous Reliability Integrity Module (ARIM) defines the structural conditions under which autonomous agent cognition remains materially stable, operationally dependable, behaviorally consistent, governance-valid, and cognitively reliable throughout changing operational conditions, runtime environments, and autonomous execution cycles.

ARIM governs autonomous reliability.

The module ensures that cognition remains operationally dependable over time and does not progressively degrade into unstable, unpredictable, inconsistent, or governance-invalid behavior.

A system satisfies ARIM only if:

- cognitive reliability remains preservable
- behavioral consistency remains maintainable
- operational stability remains demonstrable

- reliability degradation remains detectable
- cognition drift remains governable
- autonomous operation remains materially dependable

An agent cognition environment that cannot preserve reliable cognition across operational conditions does not satisfy ARIM.

Module Operational Role

ARIM defines the reliability-governance layer of ACIS.

Its role is to preserve dependable cognition throughout autonomous operation.

Module Operational Space

ARIM governs:

- autonomous reliability
- cognition stability
- behavioral consistency
- reliability degradation
- cognition drift
- operational dependability
- reliability governance
- autonomous stability

The module applies wherever autonomous cognition operates continuously or repeatedly across operational environments.

Module Function

The module applies wherever systems must preserve:

- dependable cognition
- stable operation
- predictable cognition behavior
- governance-valid reliability
- operational consistency
- autonomous resilience

Its function is to ensure that autonomous cognition remains materially reliable rather than merely occasionally successful.

Minimum Implementation Framework

1. Define the Reliability Object

The organization must define which cognition activities require reliability governance.

This may include:

- autonomous reasoning
 - planning processes
 - workflow execution
 - tool-mediated cognition
 - decision formation
 - delegation activities
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- multi-agent coordination
- continuous operational cognition

2. Define Reliability Conditions

The system must define the conditions under which autonomous reliability remains valid.

This includes:

- consistency requirements
- stability requirements
- reliability thresholds
- performance-continuity requirements
- cognition-validity requirements
- operational-dependability requirements

3. Define Reliability Degradation Detection Logic

The system must define how reliability degradation is identified.

This may include:

- cognition instability indicators
- behavioral inconsistency detection
- reliability drift indicators
- operational anomaly detection
- repeated cognition failures
- governance-validity degradation

4. Define Operational Response or Governance Logic

The system must define governance logic for reliability degradation conditions.

Governance response may include:

- reliability review
- cognition stabilization
- operational intervention
- escalation
- execution limitation
- governance reassessment
- operational invalidation where required

5. Preserve Traceability & Restrict Invalid Conditions

The system must preserve reconstructable traceability of:

- reliability assessments
- degradation events
- intervention actions
- cognition reviews
- governance decisions
- operational outcomes

An autonomous cognition system must not remain reliability-valid if cognition becomes materially unstable, inconsistent, or operationally unpredictable.

Use Case 1 — Enterprise Autonomous Operations Network

Scenario

An organization operates hundreds of autonomous agents coordinating workflows, decisions, approvals, and operational activities across multiple business environments.

Application

ARIM governs cognition stability, reliability continuity, and behavioral consistency throughout long-term autonomous operation.

Result

The organization gains stronger operational dependability, reduced cognition drift, and improved governance confidence.

Use Case 2 — Long-Term Autonomous Agent Ecosystem

Scenario

A distributed ecosystem of autonomous agents continuously interacts across planning, execution, optimization, and coordination activities.

Application

ARIM governs whether cognition remains materially reliable despite changing operational conditions and evolving execution contexts.

Result

The environment gains stronger autonomous resilience, improved predictability, and reduced exposure to cognition instability.

Canonical Closing Statement

Autonomous Reliability Integrity Module (ARIM) defines the structural conditions under which autonomous agent cognition remains materially stable, operationally dependable, behaviorally consistent, governance-valid, and cognitively reliable throughout changing operational conditions, runtime environments, and autonomous execution cycles.

Autonomous cognition cannot remain valid if it cannot remain dependable.
Autonomous reliability therefore becomes a foundational integrity condition of governable agent cognition.

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